Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

ADBMS Coursework Submission

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-linu compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

to see what all colums needed form 'pg_stats'

```
\d pg_stats;
```

to see histogram boundary values and other relevent stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

PSQL uses an Equi Depth Histogram combined with a Most Common Values list. In an equidepth histogram, all buckets contain approximately the same number of rows ($approx N/B$), while bucket boundaries vary in width according to data distribution

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)
- extra - to see the defualt number of buckets

```
SHOW default_statistics_target;
```

3.1.4 4. How Textual Attribute `title` is Put into Histogram

textual attribute "title" is put into histogram as:

1. sorting the text strings in lexicographical order like bannana and apple will be apple, bannana after sorting

2. Sampling string values from the table like table title has around 2.5M rows , it will only sample around 30K

3. Encoding initial string prefix bytes into floating-point numbers in $[0.0, 1.0]$ range for quantitative interpolation like laptop will be broken to lap with l = 108, a=97, p=112 then by 256 as between 0 and 1 , lap will be around 0.42 , we can then see where a key belongs between two keeys

4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equi-depth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From `pg_stats`)

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`id`)	Bucket Interval
B000 (Start)	66	Start of table data
B001	26031	Bucket 1: [66 - 26031]
B002	49984	Bucket 2: [26031 - 49984]
B003	74833	Bucket 3: [49984 - 74833]
B004	101140	Bucket 4: [74833 - 101140]
B005	127437	Bucket 5: [101140 - 127437]
B006	153864	Bucket 6: [127437 - 153864]

B007	179570	Bucket 7: [153864 - 179570]
B008	203627	Bucket 8: [179570 - 203627]
B009	227342	Bucket 9: [203627 - 227342]
B010	249558	Bucket 10: [227342 - 249558]
B011	274050	Bucket 11: [249558 - 274050]
B012	300161	Bucket 12: [274050 - 300161]
B013	326810	Bucket 13: [300161 - 326810]
B014	353628	Bucket 14: [326810 - 353628]
B015	378794	Bucket 15: [353628 - 378794]
B016	401971	Bucket 16: [378794 - 401971]
B017	426990	Bucket 17: [401971 - 426990]
B018	453094	Bucket 18: [426990 - 453094]
B019	480643	Bucket 19: [453094 - 480643]
B020	505273	Bucket 20: [480643 - 505273]
B021	531162	Bucket 21: [505273 - 531162]
B022	559018	Bucket 22: [531162 - 559018]
B023	582792	Bucket 23: [559018 - 582792]
B024	605800	Bucket 24: [582792 - 605800]
B025	632265	Bucket 25: [605800 - 632265]
B026	656668	Bucket 26: [632265 - 656668]
B027	683413	Bucket 27: [656668 - 683413]
B028	708726	Bucket 28: [683413 - 708726]
B029	732854	Bucket 29: [708726 - 732854]
B030	759074	Bucket 30: [732854 - 759074]
B031	784282	Bucket 31: [759074 - 784282]
B032	812447	Bucket 32: [784282 - 812447]
B033	836574	Bucket 33: [812447 - 836574]
B034	863283	Bucket 34: [836574 - 863283]
B035	888391	Bucket 35: [863283 - 888391]
B036	914667	Bucket 36: [888391 - 914667]
B037	939512	Bucket 37: [914667 - 939512]
B038	962926	Bucket 38: [939512 - 962926]
B039	987759	Bucket 39: [962926 - 987759]
B040	1013055	Bucket 40: [987759 - 1013055]
B041	1039398	Bucket 41: [1013055 - 1039398]
B042	1062735	Bucket 42: [1039398 - 1062735]
B043	1090405	Bucket 43: [1062735 - 1090405]
B044	1116673	Bucket 44: [1090405 - 1116673]
B045	1142039	Bucket 45: [1116673 - 1142039]
B046	1163912	Bucket 46: [1142039 - 1163912]
B047	1192171	Bucket 47: [1163912 - 1192171]
B048	1215031	Bucket 48: [1192171 - 1215031]
B049	1241261	Bucket 49: [1215031 - 1241261]
B050	1266561	Bucket 50: [1241261 - 1266561]
B051	1291888	Bucket 51: [1266561 - 1291888]
B052	1318652	Bucket 52: [1291888 - 1318652]
B053	1343336	Bucket 53: [1318652 - 1343336]
B054	1369813	Bucket 54: [1343336 - 1369813]
B055	1398034	Bucket 55: [1369813 - 1398034]
B056	1422999	Bucket 56: [1398034 - 1422999]
B057	1449897	Bucket 57: [1422999 - 1449897]
B058	1474101	Bucket 58: [1449897 - 1474101]
B059	1497897	Bucket 59: [1474101 - 1497897]
B060	1522641	Bucket 60: [1497897 - 1522641]
B061	1546054	Bucket 61: [1522641 - 1546054]
B062	1569842	Bucket 62: [1546054 - 1569842]
B063	1597944	Bucket 63: [1569842 - 1597944]
B064	1624272	Bucket 64: [1597944 - 1624272]
B065	1649906	Bucket 65: [1624272 - 1649906]
B066	1677958	Bucket 66: [1649906 - 1677958]

B067	1702991	Bucket 67: [1677958 - 1702991]
B068	1726251	Bucket 68: [1702991 - 1726251]
B069	1752554	Bucket 69: [1726251 - 1752554]
B070	1778916	Bucket 70: [1752554 - 1778916]
B071	1807938	Bucket 71: [1778916 - 1807938]
B072	1832794	Bucket 72: [1807938 - 1832794]
B073	1857494	Bucket 73: [1832794 - 1857494]
B074	1883037	Bucket 74: [1857494 - 1883037]
B075	1909532	Bucket 75: [1883037 - 1909532]
B076	1934685	Bucket 76: [1909532 - 1934685]
B077	1958564	Bucket 77: [1934685 - 1958564]
B078	1982713	Bucket 78: [1958564 - 1982713]
B079	2006262	Bucket 79: [1982713 - 2006262]
B080	2030390	Bucket 80: [2006262 - 2030390]
B081	2056098	Bucket 81: [2030390 - 2056098]
B082	2081337	Bucket 82: [2056098 - 2081337]
B083	2105688	Bucket 83: [2081337 - 2105688]
B084	2129827	Bucket 84: [2105688 - 2129827]
B085	2152469	Bucket 85: [2129827 - 2152469]
B086	2176821	Bucket 86: [2152469 - 2176821]
B087	2200577	Bucket 87: [2176821 - 2200577]
B088	2228756	Bucket 88: [2200577 - 2228756]
B089	2252971	Bucket 89: [2228756 - 2252971]
B090	2278837	Bucket 90: [2252971 - 2278837]
B091	2303457	Bucket 91: [2278837 - 2303457]
B092	2329311	Bucket 92: [2303457 - 2329311]
B093	2354302	Bucket 93: [2329311 - 2354302]
B094	2378735	Bucket 94: [2354302 - 2378735]
B095	2405535	Bucket 95: [2378735 - 2405535]
B096	2430421	Bucket 96: [2405535 - 2430421]
B097	2453702	Bucket 97: [2430421 - 2453702]
B098	2478306	Bucket 98: [2453702 - 2478306]
B099	2504204	Bucket 99: [2478306 - 2504204]
B100	2528285	Bucket 100: [2504204 - 2528285]

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	`\$ucces Part One`	Start of text data
B001	`(#1.1299)`	Bucket 1: [`\$ucces Part One` - `(#1.1299)`]
B002	`(#12.141)`	Bucket 2: [`(#1.1299)` - `(#12.141)`]
B003	`(#1.3792)`	Bucket 3: [`(#12.141)` - `(#1.3792)`]
B004	`(#1.56)`	Bucket 4: [`(#1.3792)` - `(#1.56)`]
B005	`(#18.11)`	Bucket 5: [`(#1.56)` - `(#18.11)`]
B006	`(1968-06-18)`	Bucket 6: [`(#18.11)` - `(1968-06-18)`]
B007	`(1982-11-08)`	Bucket 7: [`(1968-06-18)` - `(1982-11-08)`]
B008	`(1993-12-27)`	Bucket 8: [`(1982-11-08)` - `(1993-12-27)`]
B009	`(1998-02-10)`	Bucket 9: [`(1993-12-27)` - `(1998-02-10)`]
B010	`(2000-03-28)`	Bucket 10: [`(1998-02-10)` - `(2000-03-28)`]
B011	`(2002-02-22)`	Bucket 11: [`(2000-03-28)` - `(2002-02-22)`]
B012	`2003 AXA Liberty Bowl`	Bucket 12: [`(2002-02-22)` - `2003 AXA Liberty Bowl`]
B013	`(2005-04-14)`	Bucket 13: [`2003 AXA Liberty Bowl` - `(2005-04-14)`]
B014	`(2006-08-15)`	Bucket 14: [`(2005-04-14)` - `(2006-08-15)`]
B015	`(2007-11-04)`	Bucket 15: [`(2006-08-15)` - `(2007-11-04)`]
B016	`(2009-04-16)`	Bucket 16: [`(2007-11-04)` - `(2009-04-16)`]
B017	`(2010-11-13)`	Bucket 17: [`(2009-04-16)` - `(2010-11-13)`]
B018	`(2012-01-09)`	Bucket 18: [`(2010-11-13)` - `(2012-01-09)`]

B019	2012 BMW 1 Series ActiveE Electric Car	Bucket 19: [(2012-01-09) - 2012 BMW 1 Series ActiveE Electric Car]
B020	(#2.33)	Bucket 20: [2012 BMW 1 Series ActiveE Electric Car - (#2.33)]
B021	(#3.32)	Bucket 21: [(#2.33) - (#3.32)]
B022	(#5.1)	Bucket 22: [(#3.32) - (#5.1)]
B023	(#7.186)	Bucket 23: [(#5.1) - (#7.186)]
B024	A Candle for St. Jude	Bucket 24: [(#7.186) - A Candle for St. Jude]
B025	Aftermath	Bucket 25: [A Candle for St. Jude - Aftermath]
B026	Alte Schule	Bucket 26: [Aftermath - Alte Schule]
B027	Anita's Africa	Bucket 27: [Alte Schule - Anita's Africa]
B028	Asik Garip	Bucket 28: [Anita's Africa - Asik Garip]
B029	A Yen for Lester	Bucket 29: [Asik Garip - A Yen for Lester]
B030	Beatmania IIDX 18: Resort Anthem	Bucket 30: [A Yen for Lester - Beatmania IIDX 18: Resort Anthem]
B031	Bill's Thud	Bucket 31: [Beatmania IIDX 18: Resort Anthem - Bill's Thud]
B032	Bôryaku	Bucket 32: [Bill's Thud - Bôryaku]
B033	Business Must Not Interfere	Bucket 33: [Bôryaku - Business Must Not Interfere]
B034	Celebrities at SEMA 2008	Bucket 34: [Business Must Not Interfere - Celebrities at SEMA 2008]
B035	Christos Chrissopoulos	Bucket 35: [Celebrities at SEMA 2008 - Christos Chrissopoulos]
B036	Company	Bucket 36: [Christos Chrissopoulos - Company]
B037	Dad's Army	Bucket 37: [Company - Dad's Army]
B038	Dear Diary	Bucket 38: [Dad's Army - Dear Diary]
B039	Der Reinfall	Bucket 39: [Dear Diary - Der Reinfall]
B040	Die Übermacht	Bucket 40: [Der Reinfall - Die Übermacht]
B041	Double Exposure	Bucket 41: [Die Übermacht - Double Exposure]
B042	Ego	Bucket 42: [Double Exposure - Ego]
B043	El zagal y las ovejas	Bucket 43: [Ego - El zagal y las ovejas]
B044	Every Day Above Ground Is a Good Day	Bucket 44: [El zagal y las ovejas - Every Day Above Ground Is a Good Day]
B045	Ferien auf dem Martinshof	Bucket 45: [Every Day Above Ground Is a Good Day - Ferien auf dem Martinshof]
B046	Forgiving Your P.E. Teacher	Bucket 46: [Ferien auf dem Martinshof - Forgiving Your P.E. Teacher]
B047	Game Over	Bucket 47: [Forgiving Your P.E. Teacher - Game Over]
B048	God and the Man	Bucket 48: [Game Over - God and the Man]
B049	Gyözelemröl gyözelemre	Bucket 49: [God and the Man - Gyözelemröl gyözelemre]
B050	Hemingway	Bucket 50: [Gyözelemröl gyözelemre - Hemingway]
B051	Homeland in War	Bucket 51: [Hemingway - Homeland in War]
B052	Ich oder Du	Bucket 52: [Homeland in War - Ich oder Du]
B053	Inheritance	Bucket 53: [Ich oder Du - Inheritance]
B054	It's a Wonderful Life	Bucket 54: [Inheritance - It's a Wonderful Life]
B055	Joe Curren	Bucket 55: [It's a Wonderful Life - Joe Curren]

B056	Karen Han	Bucket 56: [Joe Curren - Karen Han]
B057	Kobe Bryant	Bucket 57: [Karen Han - Kobe Bryant]
B058	La garçonne	Bucket 58: [Kobe Bryant - La garçonne]
B059	La terre étrangère	Bucket 59: [La garçonne - La terre étrangère]
B060	Les étrangers	Bucket 60: [La terre étrangère - Les étrangers]
B061	Little Rock: The Politics of Murder	Bucket 61: [Les étrangers - Little Rock: The Politics of Murder]
B062	Lucka 21	Bucket 62: [Little Rock: The Politics of Murder - Lucka 21]
B063	Mar adentro	Bucket 63: [Lucka 21 - Mar adentro]
B064	Memoirs	Bucket 64: [Mar adentro - Memoirs]
B065	Miss Shellagh's Miniskirt	Bucket 65: [Memoirs - Miss Shellagh's Miniskirt]
B066	Mr. Lincoln's Springfield	Bucket 66: [Miss Shellagh's Miniskirt - Mr. Lincoln's Springfield]
B067	Napfény a jégen	Bucket 67: [Mr. Lincoln's Springfield - Napfény a jégen]
B068	Një natë pa dritë	Bucket 68: [Napfény a jégen - Një natë pa dritë]
B069	Odcinek 35	Bucket 69: [Një natë pa dritë - Odcinek 35]
B070	Operation Moonshine	Bucket 70: [Odcinek 35 - Operation Moonshine]
B071	Panic Room	Bucket 71: [Operation Moonshine - Panic Room]
B072	Pgisha Iveret	Bucket 72: [Panic Room - Pgisha Iveret]
B073	Postia	Bucket 73: [Pgisha Iveret - Postia]
B074	Que a paz sexa con vosco	Bucket 74: [Postia - Que a paz sexa con vosco]
B075	Relieved of Command/Belafonte/Who Poisoned the Children?	Bucket 75: [Que a paz sexa con vosco - Relieved of Command/Belafonte/Who Poisoned the Children?]
B076	Roll Call	Bucket 76: [Relieved of Command/Belafonte/Who Poisoned the Children? - Roll Call]
B077	Sammie Johnsin in Mexico	Bucket 77: [Roll Call - Sammie Johnsin in Mexico]
B078	Seizure Dogs	Bucket 78: [Sammie Johnsin in Mexico - Seizure Dogs]
B079	Show #486 - Wie geht es Speedy? & Playmobil Don Giovanni	Bucket 79: [Seizure Dogs - Show #486 - Wie geht es Speedy? & Playmobil Don Giovanni]
B080	Snow Buddies/Aristocats	Bucket 80: [Show #486 - Wie geht es Speedy? & Playmobil Don Giovanni - Snow Buddies/Aristocats]
B081	Spooky Spots	Bucket 81: [Snow Buddies/Aristocats - Spooky Spots]
B082	Sucedió en el internado	Bucket 82: [Spooky Spots - Sucedió en el internado]
B083	Targets	Bucket 83: [Sucedió en el internado - Targets]
B084	The Amazing Race Philippines	Bucket 84: [Targets - The Amazing Race Philippines]
B085	The Case of the Purple Pool	Bucket 85: [The Amazing Race Philippines - The Case of the Purple Pool]
B086	The Education of Shelby Knox	Bucket 86: [The Case of the Purple Pool - The Education of Shelby Knox]

B087	The Grudge	Bucket 87: [The Education of Shelby Knox - The Grudge]
B088	The Light in Her Eyes	Bucket 88: [The Grudge - The Light in Her Eyes]
B089	The Oddball Parade	Bucket 89: [The Light in Her Eyes - The Oddball Parade]
B090	The Revenger	Bucket 90: [The Oddball Parade - The Revenger]
B091	The Take Out	Bucket 91: [The Revenger - The Take Out]
B092	They are Five	Bucket 92: [The Take Out - They are Five]
B093	Tomodachi	Bucket 93: [They are Five - Tomodachi]
B094	True Anal Stories 23	Bucket 94: [Tomodachi - True Anal Stories 23]
B095	Un esprit clairvoyant	Bucket 95: [True Anal Stories 23 - Un esprit clairvoyant]
B096	Very Bad Men	Bucket 96: [Un esprit clairvoyant - Very Bad Men]
B097	Watashi no kokoro wa papa no mono	Bucket 97: [Very Bad Men - Watashi no kokoro wa papa no mono]
B098	White Wedding	Bucket 98: [Watashi no kokoro wa papa no mono - White Wedding]
B099	Yama ga aru kara..... Part 5	Bucket 99: [White Wedding - Yama ga aru kara..... Part 5]
B100	Þorsteinn Pálsson	Bucket 100: [Yama ga aru kara..... Part 5 - Þorsteinn Pálsson]

3.1.6 6. Histogram Visualization

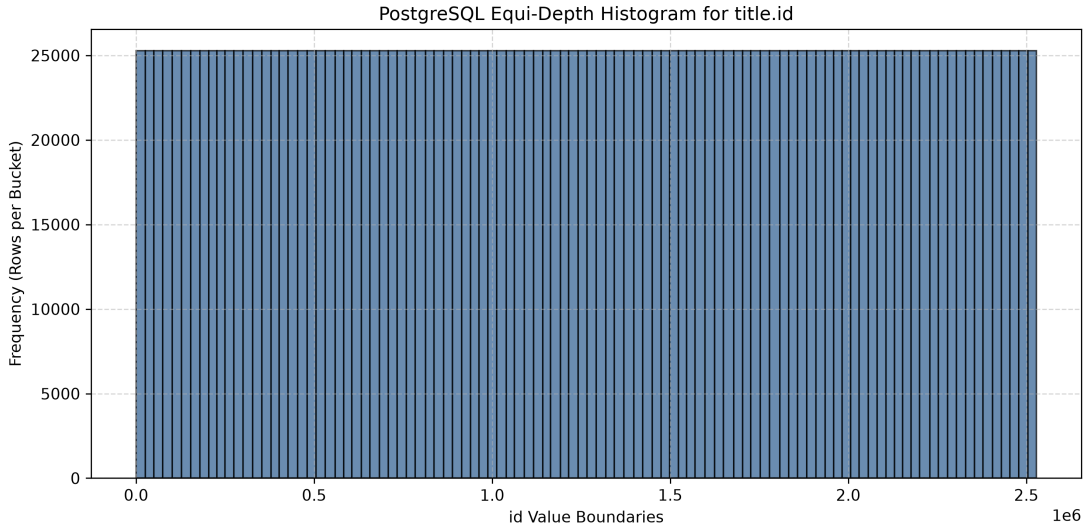


Figure 1: Histogram for title.id

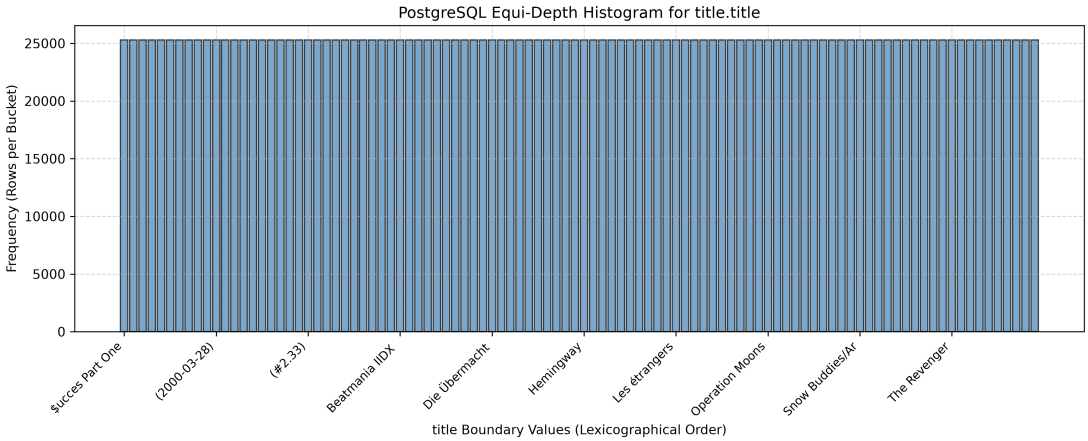


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error

- **Max Selectivity Error**: Since `default_statistics_target` is 100, there are 100 buckets. Therefore, the maximum selectivity error for a range query is bounded by:

$$\text{Max Error} \leq \frac{1}{B} = \frac{1}{100} = 0.01 \quad (1.0\%)$$

For evaluated queries(q9), the observed maximum selectivity error is **0.8031%** (0.008031), which strictly satisfies Max Error $\leq 1.0\%$.

3.1.8 8: Selectivity Estimation for Text Columns

Selectivity estimate is done for range query on text column such as "title" in PSQL such as:

*The engine scans the `histogram_bounds` array stored under pg_stats to locate the specific bucket where the target string is, after that it counts all complete buckets preceding the target bucket. Each full bucket represents a fixed fraction of total rows (1/B). For the target bucket, PostgreSQL converts initial string bytes into floating-point numbers in $[0.0, 1.0]$ range for interpolation as explain in part 4, assuming uniform distribution inside the target bucket, it calculates the target's position using:

$$\text{Fraction} = \frac{\text{Query Value} - \text{Bucket Min}}{\text{Bucket Max} - \text{Bucket Min}}$$

- Final Calculation is done by adding the fractional bucket estimate to the sum of preceding full buckets to get the final estimated selectivity.

3.1.9 9: Estimated vs. Actual Selectivities

To calculate estimated and actual selectivities of the queries and moreover there accuracy we compare estimated rows (from `EXPLAIN`) against actual rows (from `EXPLAIN ANALYZE`).

Total row count (N) for the `title` table is 2,528,312.

Query 1: `title < 'Race'`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.title < 'Race';
```

- **Actual Selectivity:** The query physically returned 1,918,141 matching rows: $S_{\text{act}} = \frac{1918141}{2528312} \approx 0.7587$ (75.87%).

```
EXPLAIN SELECT * FROM title WHERE title.title < 'Race';
```

- **Estimated Selectivity:** The planner estimated 1,938,445 rows: $S_{\text{est}} = \frac{1938445}{2528312} \approx 0.7667$ (76.67%).

- **Observation:** The absolute error is roughly 0.80%.

Query 2: `id < 350000`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.id < '350000';
```

- **Actual Selectivity:** The query returned 349,999 matching rows: $S_{\text{act}} = \frac{349999}{2528312} \approx 0.1384$ (13.84%).

```
EXPLAIN SELECT * FROM title WHERE title.id < '350000';
```

- **Estimated Selectivity:** The planner estimated 350,549 rows: $S_{\text{est}} = \frac{350549}{2528312} \approx 0.1386$ (13.86%).
- **Observation:** The absolute error here is about 0.02%.

3.2 Q2. Optimal Serial Histogram Construction ($B = 10$ Buckets)

3.2.1 1. Histogram Visualizations for Exact $N = 3000$ Samples (Slide-9 Upright Vertical Block Plots)

As defined in lecture Slide 9, a Serial Histogram sorts domain values by their ordered frequencies ($f(v_{(1)}) \leq f(v_{(2)}) \leq \cdots \leq f(v_{(m)})$) and partitions the frequency space to minimize frequency variance $\sum n_i V_i$. The upright vertical block plots below feature Frequencies on the X-axis (ordered ascending) and Value Set / Stacked Domain Elements on the Y-axis:

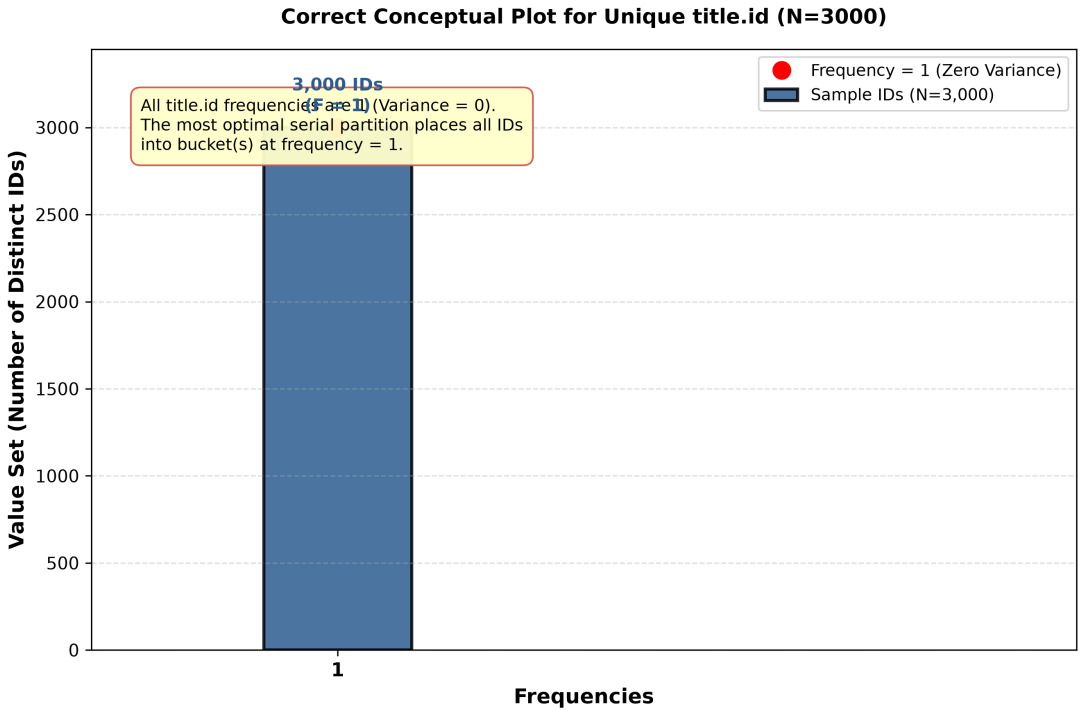


Figure 3: Correct Conceptual Plot for Unique title.id (N=3000)

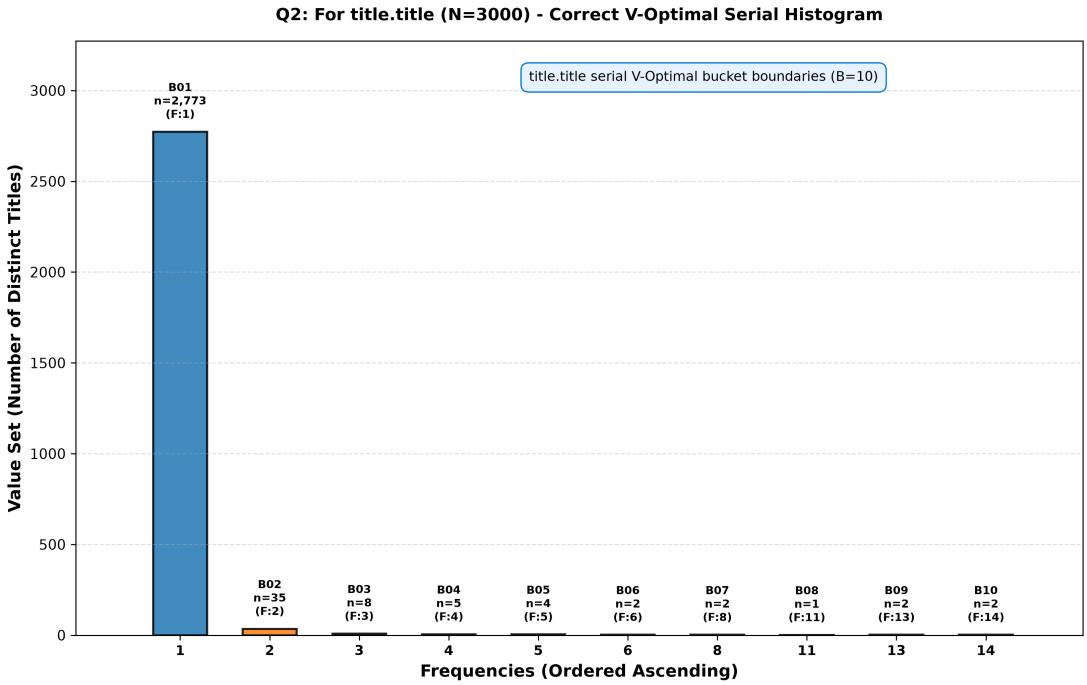


Figure 4: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

3.2.2 2. Execution Build Time (Average over 3 Runs)

- **Average Build Time for $N = 3000$: 0.0017 seconds **(`id` column)** / 0.0062 seconds** (`title` column)

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

The optimal serial histogram algorithm partitions the exact 3,000 samples into dynamic frequency buckets ($B = 10$). The exact bucket value ranges, distinct domain value counts (n_i), and frequency distributions for both columns are listed below:

A. Column `id` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

Bucket #	Value Range (`id`)	Distinct Values (n_i)	Frequency Range	Average Frequency (arf_i)
Bucket 01	[1,234 - 2,528,274]	3000	1	1.00

B. Column `title` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

Bucket #	Value Range (`title`)	Distinct Values (n_i)	Frequency Range	Average Frequency (arf_i)
Bucket 01	["\$#!+ Happens" ... "Ônna no kaôyaku"]	2773	1	1.00
Bucket 02	["(#1.132)" ... "The Box"]	35	2	2.00
Bucket 03	["(#1.118)" ... "(#4.1)"]	8	3	3.00
Bucket 04	["(#1.11)" ... "(#3.1)"]	5	4	4.00
Bucket 05	["(#1.10)" ... "Pilot"]	4	5	5.00

Bucket 06	["(#1.7)" ... "(#1.9)"]	2	6	6.00
Bucket 07	["(#1.6)" ... "(#1.8)"]	2	8	8.00
Bucket 08	["(#1.5)" ... "(#1.5)"]	1	11	11.00
Bucket 09	["(#1.1)" ... "(#1.2)"]	2	13	13.00
Bucket 10	["(#1.3)" ... "(#1.4)"]	2	14	14.00

3.2.4 4. 3-Run Selectivity Error Calculation & Average ($N = 3000$)

As required to account for sample randomness, the selectivity error evaluation was executed across **3 independent runs** (with exact sample size $N = 3000$). All three run errors are listed below along with their computed average:

A. 3-Run Max Selectivity Error Summary ($N = 3000$)

Column	Run 1 Max Error	Run 2 Max Error	Run 3 Max Error	Average Max Selectivity Error
id	0.022444	0.009556	0.009889	0.013963 (1.3963%)
title	0.612556	0.609000	0.616667	0.612741 (61.2741%)

B. Quantile-by-Quantile Error Evaluation Table for Run 1 ($N = 3000, B = 10$)

Index (k)	Query Threshold (q_k)	Actual Count (A)	Actual Selectivity (S_{act})	Estimated Count (E)	Estimated Selectivity (S_{est})	Absolute Error $\lvert S_{\text{est}} - S_{\text{act}} \rvert$
q_0	1,234.0	1	0.000333	0.00	0.000000	0.000333
q_1	282,016.2	352	0.117333	333.33	0.111111	0.006222
q_2	562,798.4	673	0.224333	666.67	0.222222	0.002111
q_3	843,580.7	966	0.322000	1000.00	0.333333	0.011333
q_4	1,124,362.9	1307	0.435667	1333.33	0.444444	0.008778
q_5	1,405,145.1	1613	0.537667	1666.67	0.555556	0.017889
q_6	1,685,927.3	1946	0.648667	2000.00	0.666667	0.018000
q_7	1,966,709.6	2266	0.755333	2333.33	0.777778	**0.022444**
q_8	2,247,491.8	2623	0.874333	2666.67	0.888889	0.014556
q_9	2,528,274.0	3000	1.000000	3000.00	1.000000	0.000000

3.2.5 5. Histograms Across Exact Sample Sizes ($N = 1000, 3000, 5000$)

Below are the optimal serial histogram visualizations ($B = 10$) for exact sample sizes $N = 1000$, $N = 3000$, and $N = 5000$:N=5000$:

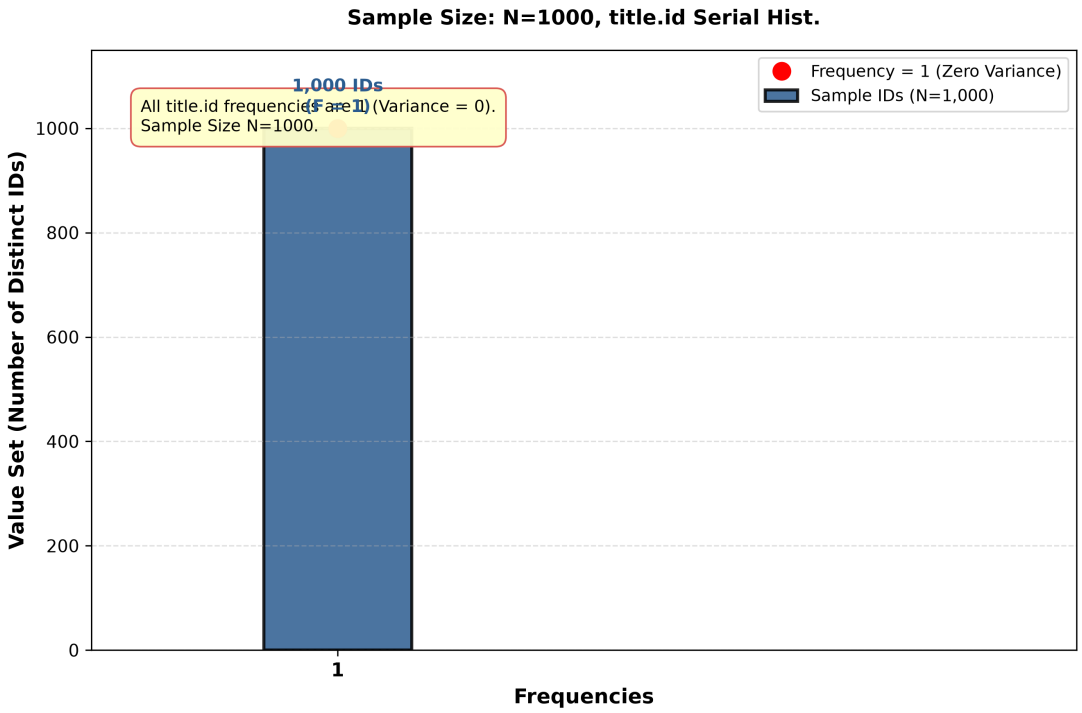


Figure 5: Sample Size: N=1000, title.id Serial Hist.

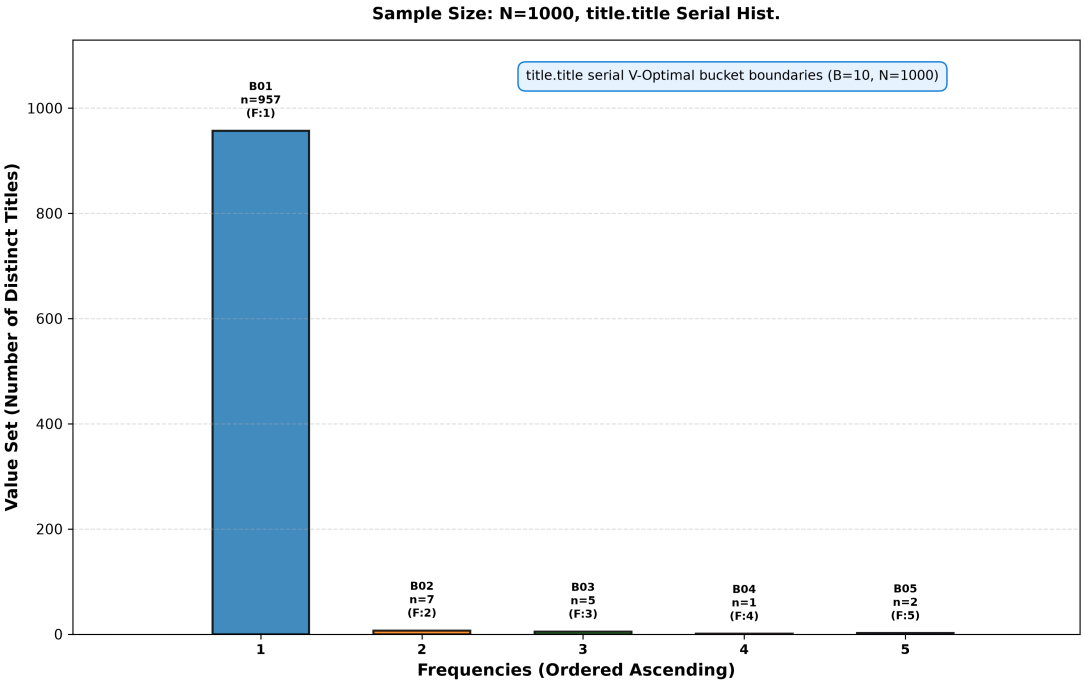


Figure 6: Sample Size: N=1000, title.title Serial Hist.

A. Sample Size $N = 1000$ Histograms

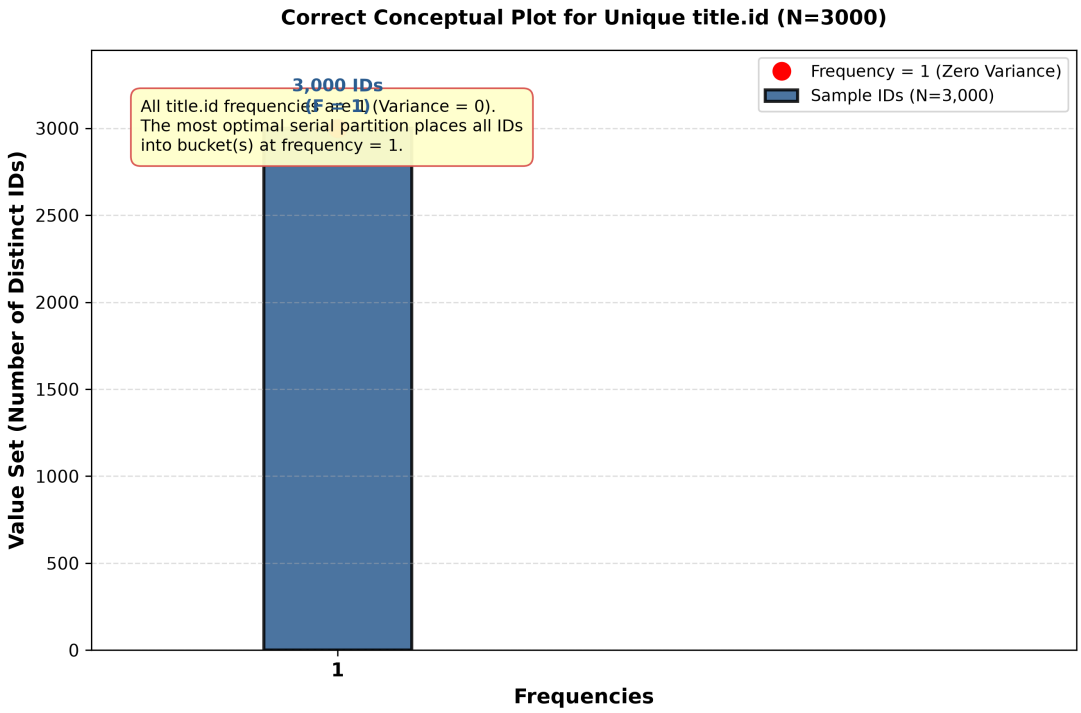


Figure 7: Correct Conceptual Plot for Unique title.id (N=3000)

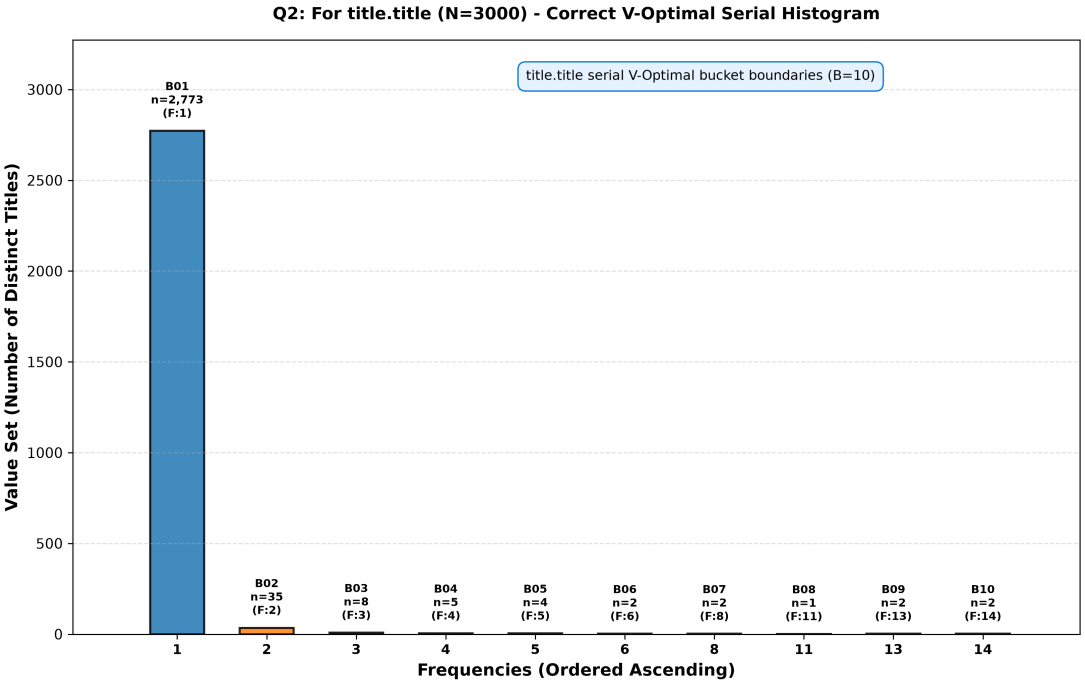


Figure 8: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

B. Sample Size $N = 3000$ Histograms

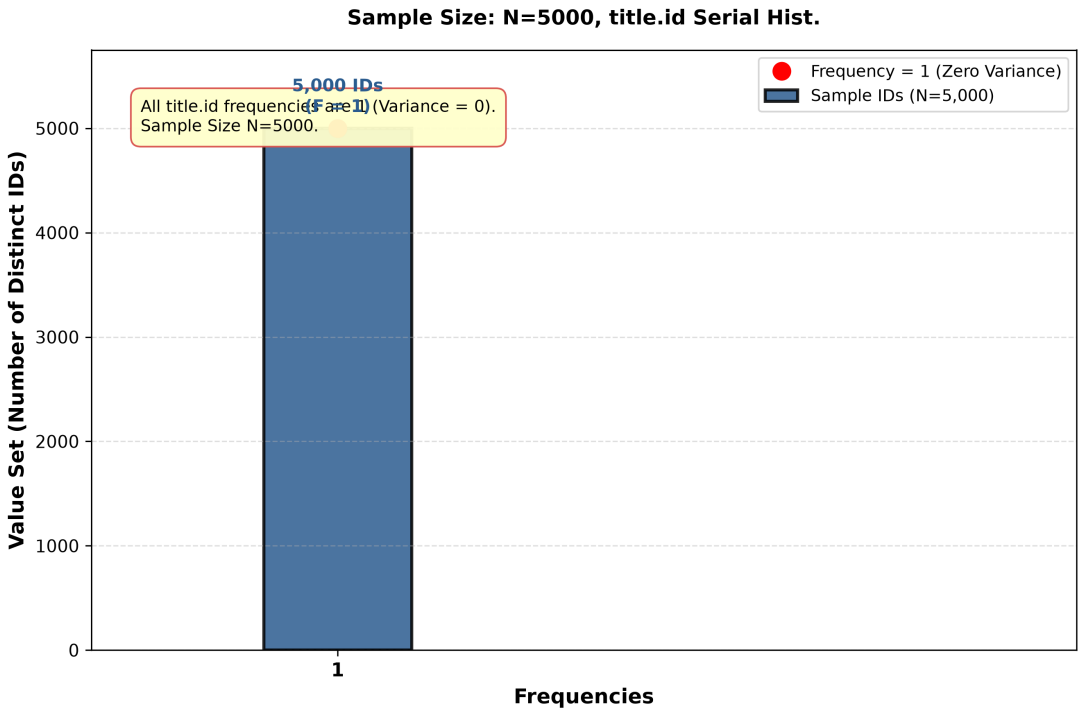


Figure 9: Sample Size: N=5000, title.id Serial Hist.

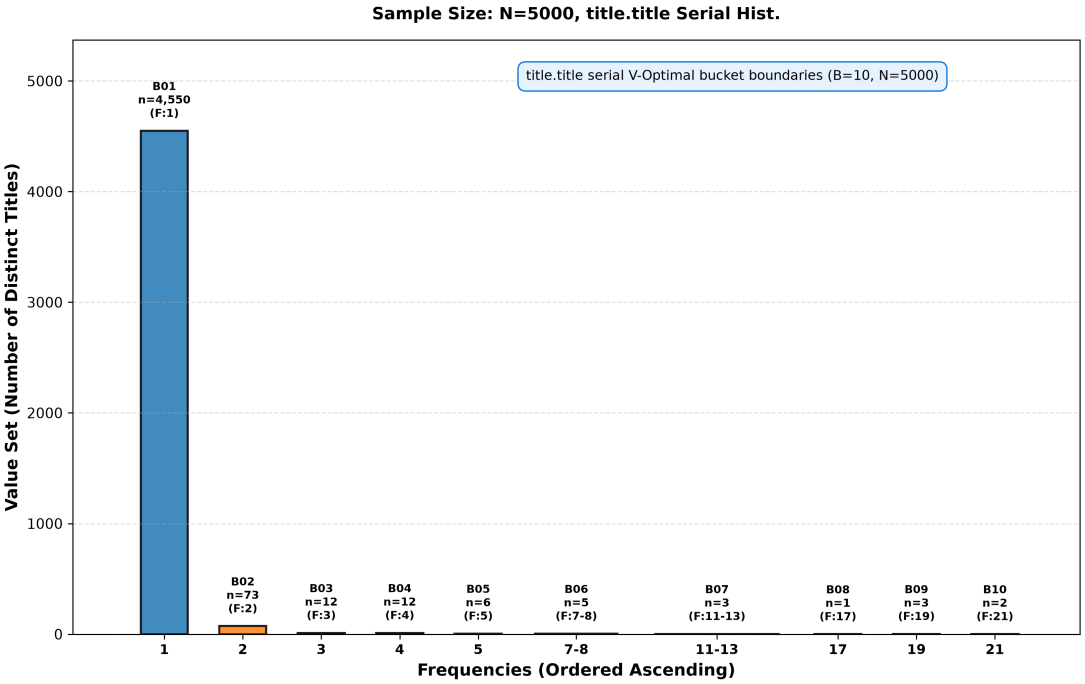


Figure 10: Sample Size: N=5000, title.title Serial Hist.

C. Sample Size $N = 5000$ Histograms

D. Comprehensive 3-Run Summary Table Across Exact Sample Sizes

Exact Sample Size (N)	Column	Run 1 Error	Run 2 Error	Run 3 Error	Average Max Error	Average Build Time
1000	id	0.020667	0.013333	0.016556	0.016852	0.0006s
1000	title	0.637000	0.625000	0.169333	0.477111	0.0013s
3000	id	0.022444	0.009556	0.009889	0.013963	0.0017s
3000	title	0.612556	0.609000	0.616667	0.612741	0.0062s
5000	id	0.013578	0.005778	0.007556	0.008970	0.0033s
5000	title	0.603667	0.600667	0.606400	0.603578	0.0080s

3.2.6 6. Full Step-by-Step Linear Regression Extrapolation ($N = 2,528,312$)

To extrapolate the construction build time for the entire `title` table ($N = 2,528,312$), we fit a linear regression model on the 3-run average measured build times:

$$T(N) = m \cdot N + c$$

A. Sample Data Points (N_i, T_i)

- Point 1: $(N_1, T_1) = (1000, 0.0006)$
- Point 2: $(N_2, T_2) = (3000, 0.0017)$
- Point 3: $(N_3, T_3) = (5000, 0.0033)$

B. Mean Calculations

- $\bar{N} = \frac{1000+3000+5000}{3} = \mathbf{3000}$
- $\bar{T} = \frac{0.0006+0.0017+0.0033}{3} = \mathbf{0.0019}$ seconds

C. Deviations & Sum of Products

N_i	T_i (s)	$(N_i - \bar{N})$	$(T_i - \bar{T})$	$(N_i - \bar{N})^2$	$(N_i - \bar{N})(T_i - \bar{T})$
1000	0.0006	-2000	-0.0013	4,000,000	2.5680
3000	0.0017	0	-0.0002	0	-0.0000
5000	0.0033	+2000	0.0014	4,000,000	2.8752
Sum				**8,000,000**	**5.4433**

D. Slope (m) and Intercept (c) Formulas

- **Slope (m)**:

$$m = \frac{\sum(N_i - \bar{N})(T_i - \bar{T})}{\sum(N_i - \bar{N})^2} = \frac{5.4433}{8,000,000} = \mathbf{0.00000068} \text{ seconds / sample}$$

- **Intercept (c)**:

$$c = \bar{T} - m \cdot \bar{N} = 0.0019 - (0.00000068 \times 3000) = -\mathbf{0.000172} \text{ seconds}$$

E. Extrapolated Build Time for Full Table ($N = 2,528,312$)

$$T(2,528,312) = (0.00000068 \times 2,528,312) + (-0.000172) = \mathbf{1.72} \text{ seconds} \quad (\mathbf{0.03} \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

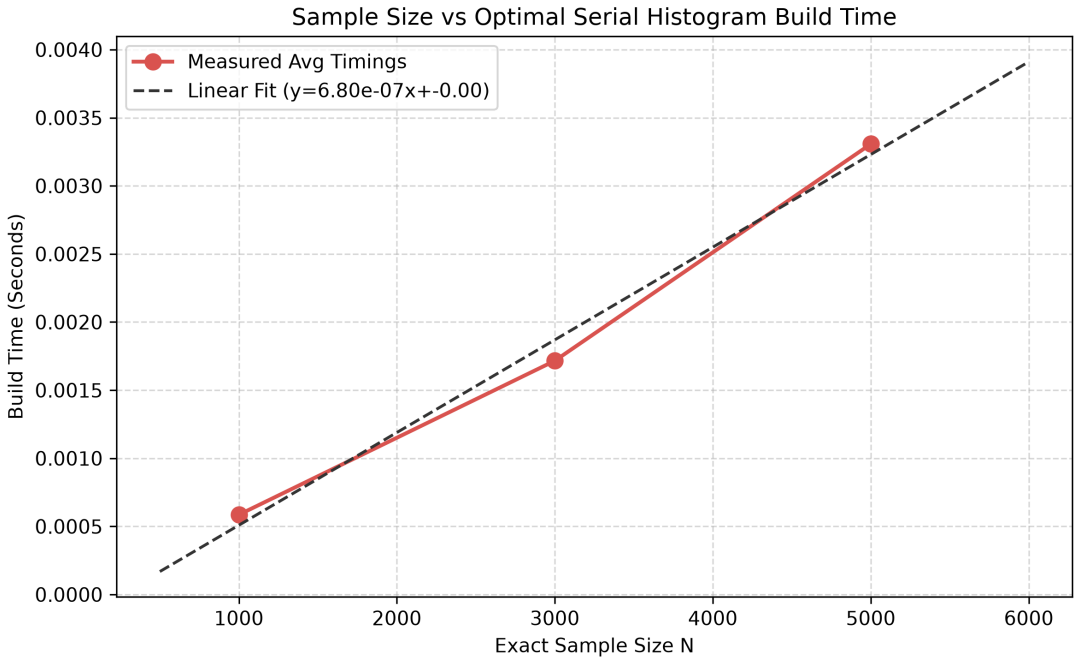


Figure 11: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

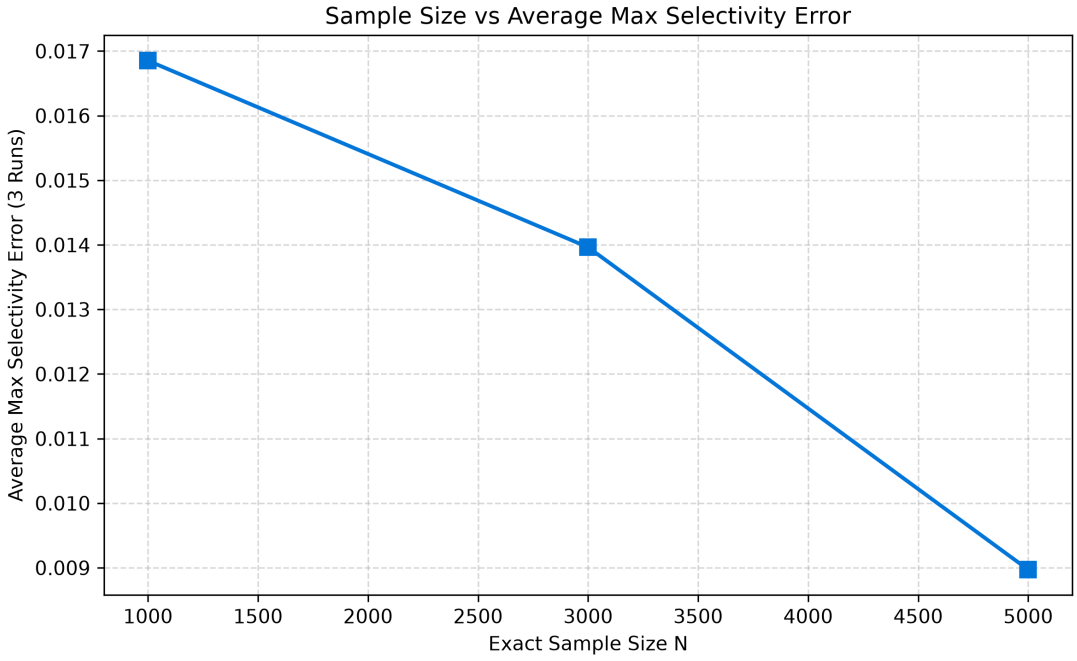


Figure 12: Sample Size vs Max Selectivity Error

3.3 9: The Trade-off of Bucket Counts

When building a histogram, choosing the number of buckets (B) is a classic trade off between performance and accuracy.

Choosing Fewer Buckets (e.g., 5 to 10)

If we use a small number of buckets, the biggest advantage is speed and efficiency. The histogram takes up very little memory in the system catalog and builds much faster because the time complexity

of our algorithm, $O(B \cdot N^2)$, drops significantly. The query planner also spends less time analyzing it. However, the major downside is that we lose a lot of data resolution. Because the buckets are so wide, the internal mathematical estimates become much less accurate, especially if the data is heavily skewed or unevenly distributed.

Choosing More Buckets (e.g., 50 to 100)

On the flip side, using a large number of buckets gives the database a high-definition view of how the data is distributed. The estimates become incredibly precise, which leads to much lower selectivity errors and helps the query optimizer make smarter decisions for complex joins. The catch is that this accuracy comes at a heavy system cost. It takes noticeably longer to calculate and construct the histogram, consumes more storage space, and adds a slight delay to the query planning phase before execution can actually begin.